

1 Deriving Continuity Equation

Beginning with a control volume, as shown below, it is possible to derive a continuity equation for mass balance.

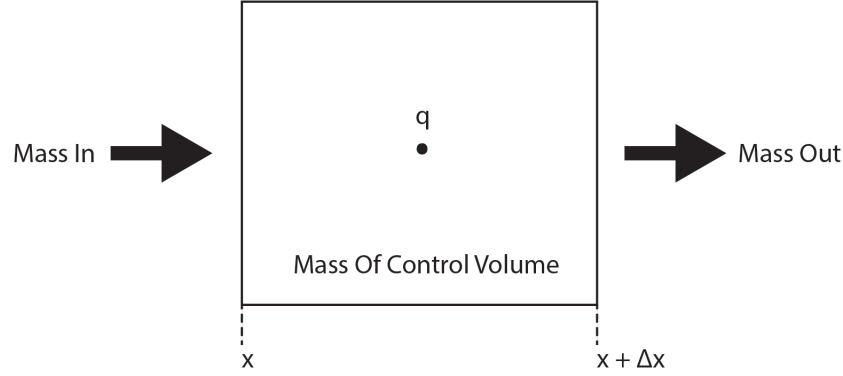


Figure 1: Control volume defined to develop mass balance equation.

Because mass can neither be created or destroyed, the mass conservation for this control volume can be described as

$$\underbrace{\text{Mass Final} - \text{Mass Initial}}_{\text{Accumulation}} = \underbrace{\text{Mass In} - \text{Mass Out}}_{\text{Convection}} + \underbrace{\text{Generation}}_{\text{Source}}, \quad (1)$$

which can be stated as

$$\underbrace{m \cdot V \Big|_{t+\Delta t} - m \cdot V \Big|_t}_{\text{Accumulation}} = \underbrace{F \cdot \Delta t \cdot A \Big|_x - F \cdot \Delta t \cdot A \Big|_{x+\Delta x}}_{\text{Convection}} + \underbrace{q \cdot \Delta t \cdot V}_{\text{Source}}. \quad (2)$$

Here, m is the mass density of the control volume, F is the mass flux, V is the volume of the control volume, Δt is the change in time, and A is the cross-sectional area orthogonal to the direction of Δx . By dividing this equation by the Δt and V , one obtains

$$\frac{m}{\Delta t} \Big|_{t+\Delta t} - \frac{m}{\Delta t} \Big|_t = \frac{F}{\Delta x} \Big|_x - \frac{F}{\Delta x} \Big|_{x+\Delta x} + q, \quad (3)$$

which can be reformulated as

$$\frac{m \Big|_{t+\Delta t} - m \Big|_t}{\Delta t} = \frac{F \Big|_x - F \Big|_{x+\Delta x}}{\Delta x} + q. \quad (4)$$

Next, the limit is taken as Δt and Δx go to zero, i.e.,

$$\lim_{\substack{\Delta t \rightarrow 0 \\ \Delta x \rightarrow 0}} \left(\frac{m \Big|_{t+\Delta t} - m \Big|_t}{\Delta t} = \frac{F \Big|_x - F \Big|_{x+\Delta x}}{\Delta x} + q \right), \quad (5)$$

resulting in

$$\frac{\partial m}{\partial t} = -\frac{\partial F}{\partial x} + q. \quad (6)$$

Note that this equation, representing the conservation of total mass, is also called the Continuity Equation. In general form, the equation can be stated as

$$\frac{\partial m}{\partial t} = -\nabla \cdot F + q. \quad (7)$$

The m (mass density of fluid per bulk volume) of the left term can be replaced like

$$\frac{\partial(\phi\rho)}{\partial t} = -\nabla \cdot F + q. \quad (8)$$

The F (mass flux in units of $\frac{kg}{m^3} \frac{m}{sec}$) can be replaced like

$$\frac{\partial(\phi\rho)}{\partial t} = -\nabla \cdot (\rho \cdot u) + q. \quad (9)$$

In this formulation, there are no source terms (wells), such that,

$$\frac{\partial(\phi\rho)}{\partial t} = -\nabla \cdot (\rho \cdot u). \quad (10)$$

Note further that the fluid is incompressible and porosity is constant, such that the accumulation term vanishes,

$$0 = -\nabla \cdot (\rho \cdot u). \quad (11)$$

Next, by assuming that fluid transport is entirely governed by pressure-driven advection (diffusion is neglected), and that fluid velocity can be described by Darcy's Law, $u = -\frac{k}{\mu} \nabla P$, the continuity equation is given by

$$-\nabla \cdot \left(\rho \left(-\frac{k}{\mu} \nabla P \right) \right) = 0, \quad (12)$$

which simplifies to,

$$\nabla \cdot \left(\rho \frac{k}{\mu} \nabla P \right) = 0, \quad (13)$$

Considering that permeability, density, and viscosity are constant,

$$\rho \frac{k}{\mu} \nabla \cdot (\nabla P) = 0. \quad (14)$$

Therefore, the continuity equation is,

$$\nabla^2 P = 0. \quad (15)$$

2 Finite Volume Method Discretization

Beginning with the continuity equation from mass balance, Equation 15,

$$\nabla^2 P = 0, \quad (16)$$

the equation is integrated over a control volume V_i ,

$$\int_{V_i} \nabla \cdot (\nabla P) dV = 0. \quad (17)$$

Applying the divergence theorem converts the volume integral into a surface integral,

$$\int_{A_i} \nabla P \cdot \mathbf{n} dA = 0, \quad (18)$$

where $\mathbf{n}$ is the outward-pointing unit normal vector.

For a one-dimensional domain, the fluxes through the left and right faces of the control volume can be approximated using finite differences,

$$A \left(\frac{P_{i+1} - P_i}{\Delta x} \right) - A \left(\frac{P_i - P_{i-1}}{\Delta x} \right) = 0, \quad (19)$$

where A is the cross-sectional area normal to the flow direction and i indicates cell number.

Rearranging yields

$$\frac{A}{\Delta x} (P_{i+1} - 2P_i + P_{i-1}) = 0. \quad (20)$$

Introducing the transmissibility,

$$T = \frac{kA}{\mu \Delta x}, \quad (21)$$

the discretized equation becomes

$$-TP_{i-1} + 2TP_i - TP_{i+1} = 0. \quad (22)$$

Note that, at the constant-pressure (Dirichlet) boundaries, the boundary transmissibility is

$$T_B = \frac{kA}{\mu \frac{\Delta x}{2}}, \quad (23)$$

and Equation 22 reads,

$$-T_B P_L + (T + T_B) P_1 - TP_2 = 0, \quad -TP_{N_x-1} + (T + T_B) P_{N_x} - T_B P_R = 0, \quad (24)$$

for the left and right boundaries, respectively. Here, N_x is the number of cells in the domain and P_L and P_R are the left and right pressure boundary conditions, respectively. These equations can be rewritten as,

$$(T + T_B) P_1 - TP_2 = T_B P_L, \quad -TP_{N_x-1} + (T + T_B) P_{N_x} = T_B P_R, \quad (25)$$

to facilitate their incorporation in the linear system of equations. Indeed, the discretized equations may be assembled into a linear system,

$$\mathbf{A}\mathbf{P} = \mathbf{q}, \quad (26)$$

where $\mathbf{A}$ is the transmissibility matrix based on Equation 22, $\mathbf{q}$ contains the contributions from the prescribed pressure boundary conditions, and where $\mathbf{P}$ is the vector of unknown cell-center pressures,

$$\begin{bmatrix} T_B + T & -T & 0 & \cdots & 0 \\ -T & 2T & -T & \cdots & 0 \\ 0 & -T & 2T & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & -T & T + T_B \end{bmatrix} \begin{bmatrix} P_1 \\ P_2 \\ P_3 \\ \vdots \\ P_{N_x} \end{bmatrix} = \begin{bmatrix} T_B P_L \\ 0 \\ 0 \\ \vdots \\ T_B P_R \end{bmatrix}. \quad (27)$$

The system of equations is then solved as,

$$\mathbf{P} = \mathbf{A}^{-1}\mathbf{q}. \quad (28)$$

The fluid velocity across the interfaces can then be found using Darcy's Law as,

$$u = -\frac{k}{\mu} \frac{\partial P}{\partial x}. \quad (29)$$

Applying a finite-difference approximation at interface $i + \frac{1}{2}$ gives

$$u_{i+\frac{1}{2}} = -\frac{k}{\mu} \left(\frac{P_{i+1} - P_i}{\Delta x} \right). \quad (30)$$

Note that for the domain boundaries these velocities are,

$$u_L = -\frac{k}{\mu} \left(\frac{P_1 - P_L}{\frac{\Delta x}{2}} \right), \quad u_R = -\frac{k}{\mu} \left(\frac{P_R - P_{N_x}}{\frac{\Delta x}{2}} \right). \quad (31)$$